Adaptive Robust Interaction Force Control of a Robotic Manipulator in Uncertain Environments

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***Abstract*—This article presents a novel adaptive robust interaction force control approach that aims to achieve accurate and robust force modulation of a robot manipulator in uncertain environments. First, a novel interaction model between manipulator and environment is developed that captures both structured and unstructured uncertainties. According to the developed interaction model, a two-loop control scheme is developed, which consists of an adaptive robust interaction controller (ARIC) in the outer loop and a motion tracking controller in the inner loop. With the proposed ARIC, the unknown structured parameters of the developed interaction model are estimated online. These estimated parameters are then utilized to generate feedforward compensation actions in the mechanism of ARIC, while unstructured uncertainties are addressed through robust feedback control so that excellent interaction force tracking accuracy can be ensured. In addition, the theoretical stability analysis of the ARIC is presented and practical experiments are carried out on a robot manipulator. The experimental results confirm the superiority of the proposed approach over these typical algorithms in terms of interaction force tracking performance in different environments.**

***Index Terms*—Adaptive robust control, interaction force control, robotic manipulator, uncertain environments.**

I. Introduction

In recent years, interaction force control has emerged as a prominent topic in robotics, which is applied in various applications such as peg-in-hole operations [1], human–robot collaboration [2], and robotic ultrasound imaging [3]. Currently, typical applications of interaction force control can generally be categorized into three types: robot–environment interaction (REI), human–robot interaction (HRI), and human–robot–environment interaction (HREI). It is known that the accurate modulation of the interaction force is a fundamental issue in REI, HRI, and HREI [4], which is the key factor determining the success or failure of a task execution.

Prominent interaction force control methodologies include impedance control (IC) [5], admittance control (AC) [6], hybrid force/position control [7], parallel force/position control [8], and others. IC and AC have particularly attracted widespread interests and are commonly referred to as force-based IC and position-based IC, respectively [9]. These techniques aim at letting a robot show a desired compliant behavior by conceptualizing the robot as a second-order mass–spring–damper system, thereby establishing the desired dynamic relationship between the interaction force and position/velocity [10]. However, for uncertain environments, traditional IC and AC are difficult to achieve accurate interaction force control. Generally, achieving a high steady-state force tracking accuracy significantly hinges on an accurate planning of the impedance controller reference trajectory [11]. However, precise trajectory planning relies on comprehensive knowledge of environment information, which is often incomplete or unknown in practical scenarios [12].The uncertainties associated with environmental information manifest in either fully unknown or partially known geometric and dynamic characteristics [13]. While geometric information, such as the location of the environment, can be imperfectly obtained through vision or laser sensors before commencing a contact interactive task, dynamic information about the environment can only be estimated after the contact interaction has occurred.

To address these challenges, numerous studies have concentrated on identifying or estimating environmental dynamics to modify the reference trajectory of the impedance/admittance controller. Comprehensive analyses of environment impedance identification and estimation approaches are presented in [14], [15]. In [16], [17], [18], the recursive least-squares (RLS) method was employed for the online identification of environmental impedance. For instance, [16] achieved a even level of accuracy in estimating environmental stiffness in the absence of persistent excitation from reference signals. However, [17] indicated that accurately estimating environmental damping relies on persistent excitation from the reference signals. They addressed the dependence on precise estimation of environmental parameters by integrating system identification

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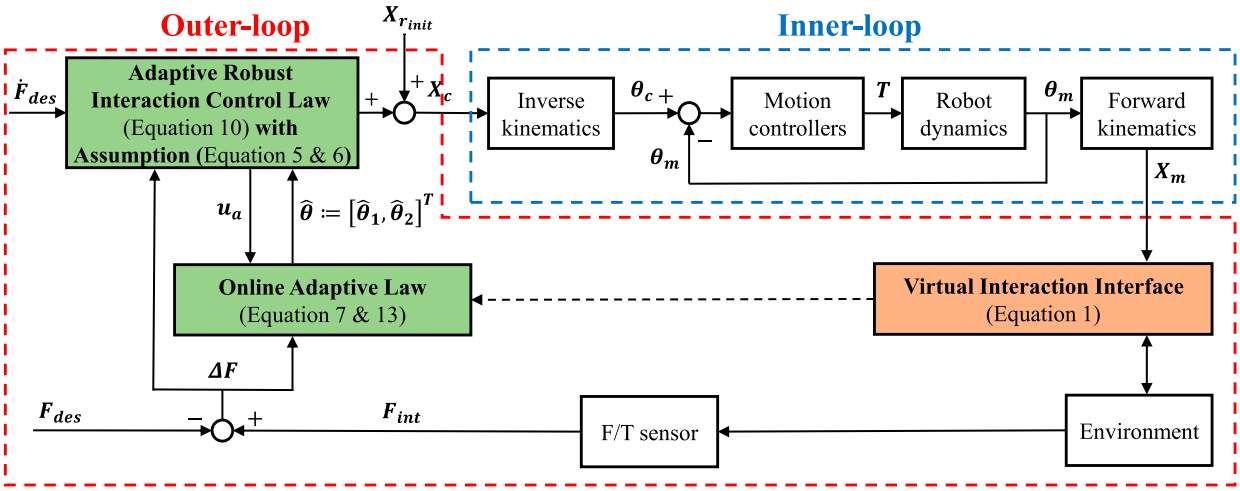


Fig. 1. Overall framework of the proposed two-loop adaptive robust interaction control scheme.

with a pre-trained offline artificial neural network. In addition, adaptive techniques are frequently employed to estimate environmental impedance [19], [20], [21]. In [19], an adaptive control algorithm was designed to estimate the position and stiffness of the contact surface, which was utilized to calculate the reference position trajectory. The force tracking performance was validated on a levitated microrobot. Seraji and Colbaugh [20] proposed two force control schemes in the absence of environmental information. The first scheme directly modifies the reference trajectory of the impedance controller using the model-reference adaptive control method, which is extremely simple and computationally efficient. The second scheme updates the reference trajectory using an indirect adaptive strategy that estimates environmental stiffness and position in real-time. It is conceptually simple and tends to be robust against unmodeled effects and sensor noise. Although the aforementioned methods can estimate environmental impedance parameters, they can only estimate partial environment information. This limitation arises because these methods typically model the environment as a nominal mass-spring-damper system when estimating parameters, with numerous studies only considering the stiffness property for the sake of simplified design [22]. The nominal mass-spring-damper and spring models have difficulties in capturing the complexities and unstructured uncertainties of environmental dynamics in various scenarios. Consequently, the impedance/admittance approaches based on reference trajectory adjustments are generally sensitive to unstructured uncertainties in interaction behaviors, which typically include modeling errors and unknown external disturbances. In addition, several studies utilize the extended Kalman filtering (EKF) technique to obtain environmental information that is used to adjust or analytically calculate the controller gain [23], [24], [25]. However, [26] pointed out that noise interference in force signals can lead to significant performance degradations in the EKF-based approaches.

In addition to the studies mentioned earlier, several variable impedance/admittance control strategies have been developed [11], [13], [27], [28]. Duan et al. [11] proposed an adaptive variable impedance control method along with a stability and convergence analysis. In this method, the impedance parameters are dynamically adjusted in real time based on the key parameter, the update rate σ, to compensate for the uncertain position of the environment. A larger σ enhances the controller's ability to adapt to environmental uncertainties. However, as σ increases, the transient overshoot in contact force tracking becomes more pronounced. Kim et al. [13] proposed a robust interaction control strategy consisting of two phases: stability enforcement (SE) and performance enforcement (PE). [13] indicates that during the PE phase, impedance/admittance adaptive techniques can be applied to improve the steady-state contact force after estimating the stiffness of the uncertain environment. To some extent, these methods contribute to mitigating the adverse effects of both robot and environmental modeling uncertainties on interaction force control. Nevertheless, these controllers may struggle to achieve precise control of interaction forces and suffer from significant force overshoots during contact transients [23]. Furthermore, IC-based approaches are susceptible to the dynamic uncertainties of the robot, as they inherently depend on accurate dynamic models [29], [30]. In contrast, AC-based methods eliminate this direct reliance on the robot's dynamic model. As a result, the robustness of interaction force control against dynamic uncertainties can be improved through the elaborate synthesis of a motion controller. Additionally, existing AC-based algorithms typically assume that the motion controller can perfectly track the commanded trajectory, i.e., $\boldsymbol{X}_m = \boldsymbol{X}_c$ in Fig. 1. This assumption is often unrealistic in practical interaction tasks, leading to a degradation in the accuracy of interaction force control [29], [30].

In addition, numerous intelligent control methods such as impedance learning [31], adaptive neural control [32] and reinforcement learning [33] are studied in details. In fact, learning-based approaches not only require a significant amount of data and involve pretraining costs, but also rely on prior knowledge of the system. Consequently, they are fundamentally unsuitable for practical contact tasks [11], [19], [31].

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Recognizing the limitations of existing interaction force control approaches, we propose a novel adaptive robust interaction force control algorithm for a robot manipulator interacting with uncertain environments. As shown in Fig. 1, the proposed approach consists of a two-loop structure. In the outer loop, we introduce a novel dynamic interaction model that effectively captures both structured and unstructured uncertainties. Structured uncertainties arise from unknown parameters, while unstructured uncertainties include inner-loop motion tracking errors, external unmeasured disturbances, and model compensation errors. With the proposed dynamic interaction model, we synthesize an adaptive robust interaction controller (ARIC) to effectively address these uncertainties. By incorporating an adaptive mechanism, structured uncertainties can be estimated online, leading to an improved feedforward compensation. Additionally, the robust feedback control law in the ARIC can effectively address unstructured uncertainties. The ARIC in the outer loop generates the commanded motion of the robot manipulator, which is then transmitted into the inner-loop motion controller. Accurate and robust interaction force control performance is guaranteed theoretically and confirmed by practical experiments. The contributions of this article are summarized as follows.

1) A novel dynamic robot–environment interaction model is proposed that captures both the structured and unstructured uncertainties of interactions.
2) With the proposed dynamic interaction model, an ARIC is synthesized. This controller estimates unknown structured parameters online and mitigates unstructured uncertainties through robust feedback.
3) In addition to the theoretical stability analysis, real-world experiments are also carried out to validate the effectiveness of the proposed controller.

The reminder of this article is organized as follows. In Section II, we introduce the proposed dynamic interaction model and present the problem formulation. Section III focuses on the synthesis of the ARIC and demonstrates the theoretical stability and convergence analysis. The experimental setup and the comparative results are presented in Section IV. Furthermore, Section V illustrates the limitations of the proposed method and outlines potential directions for future improvements. Finally, the conclusion is drawn in Section VI.

II. Dynamic Interaction Model and Problem Formulation

This section develops a novel dynamic interaction model between the robot and environment, which is then employed to synthesize an ARIC in the following section.

Different from the traditional static interaction models used in impedance/admittance control approaches, a novel dynamic interaction model that captures both structured and unstructured uncertainties of the interaction process is proposed. This model is employed in the controller deign, resulting in a function describing the relationship between the interaction force and the motion of the robot and environment. The proposed interaction model is formulated as follows:

$$\boldsymbol{F}_{\mathbf{int}} = \int_0^t \boldsymbol{K}(\boldsymbol{X_e} - \boldsymbol{X_m})dt + \tilde{\boldsymbol{\Delta}}_{\mathbf{10}} \tag{1}$$

where $\boldsymbol{F}_{\mathbf{int}} \in \mathbb{R}^6$ represents the interaction force/torque in the Cartesian space, which can be measured in real-time by the six-dimensional force/torque sensor mounted on the robot's end-effector. $\boldsymbol{K} \in \mathbb{R}^{6\times 6}$ is a positive-definite matrix without any special physical meaning. $\tilde{\boldsymbol{\Delta}}_{\mathbf{10}} \in \mathbb{R}^6$ represents the unstructured uncertainties, $\boldsymbol{X_e} \in \mathbb{R}^6$ and $\boldsymbol{X_m} \in \mathbb{R}^6$ denote the position of the environment and the end-effector, respectively. In (1), $\boldsymbol{K}$ is unknown and referred to structured uncertainties. Differentiating (1) yields

$$\frac{d}{dt}\boldsymbol{F}_{\mathbf{int}} = \boldsymbol{K}(\boldsymbol{X_e} - \boldsymbol{X_m}) + \tilde{\boldsymbol{\Delta}}_{\mathbf{1}} \tag{2}$$

where $\tilde{\boldsymbol{\Delta}}_{\mathbf{1}} = (d/dt)\tilde{\boldsymbol{\Delta}}_{\mathbf{10}}$. For the sake of illustrating the controller design and analysis in a more simple and clear way, one dimension of (2) is investigated here

$$\frac{d}{dt} f_{\text{int}} = k(x_e - x_m) + \tilde{\delta}_1. \tag{3}$$

Let f_{des} be the desired interaction force. The control objective of this article is to synthesize a motion law x_m for the robotic manipulator so that the actual interaction force f_{int} approximates f_{des} as close as possible regardless of those structured and unstructured uncertainties in the interaction process.

III. Synthesis of the ARIC

A. ARIC Law

To synthesize the motion law x_m, (3) is rewritten as the following state-space form

$$\dot{x} = -\theta_1 u + \theta_2 + \tilde{\delta}_1 \tag{4}$$

where the system state $x = f_{\text{int}}$, the control input $u = x_m$, the unknown structured parameters $\boldsymbol{\theta} := [\theta_1, \theta_2]^T = [k, kx_e]^T$. The following assumption which is beneficial to the controller design is made.

Assumption: The unknown structured parameter $\boldsymbol{\theta}$ and the unstructured uncertainty $\tilde{\delta}_1$ are bounded, as detailed in the following equations:

$$\boldsymbol{\theta} \in \Omega_\theta \triangleq \{\boldsymbol{\theta} : \boldsymbol{\theta}_{\mathbf{min}} \le \boldsymbol{\theta} \le \boldsymbol{\theta}_{\mathbf{max}}\} \tag{5}$$

$$\tilde{\delta}_1 \in \Omega_{\tilde{\delta}_1} \triangleq \left\{\tilde{\delta}_1 : |\tilde{\delta}_1| \le \delta_{\max}\right\} \tag{6}$$

where $\boldsymbol{\theta}_{\mathbf{max}} = [\theta_{1\max}, \theta_{2\max}]^T$, $\boldsymbol{\theta}_{\mathbf{min}} = [\theta_{1\min}, \theta_{2\min}]^T$.

Define $\tilde{\boldsymbol{\theta}} := \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}$ as the parameter estimation error, $\hat{\boldsymbol{\theta}}$ denotes the estimated value of $\boldsymbol{\theta}$. The following discontinuous projection-based adaptive law is synthesized

$$\dot{\hat{\boldsymbol{\theta}}} = \text{Proj}_{\hat{\boldsymbol{\theta}}}(\boldsymbol{\Gamma}\boldsymbol{\tau}) \tag{7}$$

where $\boldsymbol{\Gamma} > \mathbf{0}$ is the adaptive gain matrix with a positive and diagonal form, $\boldsymbol{\tau}$ is the adaptation function and will be designed

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later. The element of the projection mapping operator in (7) is defined as

$$\text{Proj}_{\hat{\theta}_i}(\bullet_i)=\begin{cases}0, & \text{if } \hat{\theta}_i=\theta_{i\max} \text{ and } \bullet_i>0\\ 0, & \text{if } \hat{\theta}_i=\theta_{i\min} \text{ and } \bullet_i<0\\ \bullet_i, & \text{otherwise.}\end{cases}. \tag{8}$$

Let $x_d=f_{\text{des}}$, then the force tracking error $z=x-x_d$. Differentiating z and considering (4) yield

$$\dot{z}=-\theta_1 u+\theta_2+\tilde{\delta}_1-\dot{x}_d. \tag{9}$$

Next, the control law u of ARIC is synthesized as follows:

$$\begin{cases}u=u_a+u_s\\ u_a=-(\dot{x}_d-\hat{\theta}_2)/\hat{\theta}_1\\ u_s=\dfrac{k_1 z}{\theta_{1\min}}+u_{msn}\end{cases} \tag{10}$$

where u_a and u_s represent the feedforward compensation term and the robust feedback term, respectively, k_1 is the positive robust feedback gain, and u_{msn} is the nonlinear feedback term satisfying

$$\begin{cases}z[-\boldsymbol{\varphi}^T\tilde{\boldsymbol{\theta}}+\tilde{\delta}_1+(-u_{msn}\theta_1)]\le\varepsilon\\ -zu_{msn}\theta_1\le 0\end{cases} \tag{11}$$

where $\boldsymbol{\varphi}=[-u_a,1]^T$ and ε is an arbitrarily small positive value. Furthermore, substituting (10) into (9), we obtain the following error dynamics

$$\dot{z}=-\frac{\theta_1}{\theta_{1\min}}(k_1 z)-\boldsymbol{\varphi}^T\tilde{\boldsymbol{\theta}}+\tilde{\delta}_1+(-u_{msn}\theta_1). \tag{12}$$

B. Theoretical Results

The adaptation function in (7) is designed as

$$\boldsymbol{\tau}=\boldsymbol{\varphi}z \tag{13}$$

which is utilized to estimate unknown structured parameters online to achieve accurate feedforward compensation. By employing this adaptation function, the proposed control law (10) of ARIC can guarantee the following theoretical results:

Theorem 1: The interaction force error will be ultimately bounded by $\sqrt{(\theta_{1\min}\varepsilon/k_1\theta_1)}$, i.e., $|z(\infty)|\le\sqrt{(\theta_{1\min}\varepsilon/k_1\theta_1)}$.

Proof 1: First, we proceed to present the proof of Theorem 1. We need to find an appropriate Lyapunov function to prove the boundedness of the interaction force tracking error. To begin with, the following Lyapunov function is defined

$$V=\frac{1}{2}z^2 \tag{14}$$

by differentiating V and applying the error dynamics (12), we obtain

$$\begin{aligned}\dot{V}&=z\dot{z}\\ &=-\frac{\theta_1}{\theta_{1\min}}(k_1 z^2)+z[-\boldsymbol{\varphi}^T\tilde{\boldsymbol{\theta}}+\tilde{\delta}_1+(-u_{msn}\theta_1)]\\ &\le-\frac{\theta_1}{\theta_{1\min}}k_1 z^2+\varepsilon\\ &=-\alpha V+\beta\end{aligned} \tag{15}$$

in which $\alpha=2(\theta_1/\theta_{1\min})k_1$, $\beta=\varepsilon$. By employing the comparison lemma, one can obtain

$$V(t)\le V(0)\exp(-\alpha t)+\frac{\beta}{\alpha}[1-\exp(-\alpha t)] \tag{16}$$

which implies $|z(\infty)|\le\sqrt{(\theta_{1\min}\varepsilon/k_1\theta_1)}$. Thus, we conclude the proof of Theorem 1.

Theorem 2: If the interaction model is free of unstructured uncertainties, i.e. $\tilde{\delta}_1=0$, the interaction force error will eventually converge to zero, i.e., $z\to 0$, as $t\to\infty$.

Proof 2: Next, we identify an enhanced Lyapunov function to prove Theorem 2. We introduce a new Lyapunov candidate function $V_a=(1/2)z^2+(1/2)\tilde{\boldsymbol{\theta}}^T\boldsymbol{\Gamma}^{-1}\tilde{\boldsymbol{\theta}}$, then differentiate it to obtain

$$\begin{aligned}\dot{V}_a&=z\dot{z}+\tilde{\boldsymbol{\theta}}^T\boldsymbol{\Gamma}^{-1}\dot{\tilde{\boldsymbol{\theta}}}\\ &=\left\{-\frac{\theta_1}{\theta_{1\min}}k_1 z^2+z[(-\boldsymbol{\varphi}^T\tilde{\boldsymbol{\theta}})+\tilde{\delta}_1+(-u_{msn}\theta_1)]\right\}\\ &\quad+\tilde{\boldsymbol{\theta}}^T\boldsymbol{\Gamma}^{-1}\text{Proj}_{\hat{\boldsymbol{\theta}}}(\boldsymbol{\Gamma}\boldsymbol{\varphi}z)\\ &=-\frac{\theta_1}{\theta_{1\min}}k_1 z^2+z(-u_{msn}\theta_1)+z\tilde{\delta}_1\\ &\quad+\tilde{\boldsymbol{\theta}}^T\boldsymbol{\Gamma}^{-1}[\text{Proj}_{\hat{\boldsymbol{\theta}}}(\boldsymbol{\Gamma}\boldsymbol{\varphi}z)-\boldsymbol{\Gamma}(\boldsymbol{\varphi}z)]\end{aligned} \tag{17}$$

substituting (11) and the second property of the discontinuous projection operator [34], i.e., $\tilde{\boldsymbol{\theta}}^T(\boldsymbol{\Gamma}^{-1}\text{Proj}_{\hat{\boldsymbol{\theta}}}(\boldsymbol{\Gamma}\boldsymbol{\tau})-\boldsymbol{\tau})\le 0\ \forall\boldsymbol{\tau}$, into (17), yields

$$\dot{V}_a\le-\frac{\theta_1}{\theta_{1\min}}k_1 z^2 \tag{18}$$

which indicates that $z\in\mathcal{L}_2$. Furthermore, it is easy to verify that $z\in\mathcal{L}_\infty$ from Theorem 1, and $\dot{z}\in\mathcal{L}_\infty$ from (9). By applying Barbalat's lemma [35], one can guarantee that $z(\infty)=0$, leading to Theorem 2.

Theorem 3: On the basis of Theorem 2, if the desired trajectory satisfies the following persistent exciting (PE) condition

$$\exists\, T,t_0,\varepsilon_p>0 \text{ s.t.}\int_{t-T}^{t}\boldsymbol{\varphi}(x_d(v))\boldsymbol{\varphi}(x_d(v))^T dv\ge\varepsilon_p I_p,\ \forall t\ge t_0 \tag{19}$$

then, the parameter estimates $\hat{\boldsymbol{\theta}}$ asymptotically converge to their true values, i.e., $\tilde{\boldsymbol{\theta}}\to 0$ when $t\to\infty$.

Proof 3: Finally, we prove the converging accuracy of the estimated parameters. Differentiating the error dynamics (12), yields

$$\ddot{z}=-\frac{\dot{\theta}_1}{\theta_{1\min}}k_1 z-\frac{\theta_1}{\theta_{1\min}}k_1\dot{z}-\dot{\boldsymbol{\varphi}}^T\tilde{\boldsymbol{\theta}}-\boldsymbol{\varphi}^T\dot{\tilde{\boldsymbol{\theta}}}-\dot{\theta}_1 u_{msn}-\theta_1\dot{u}_{msn}. \tag{20}$$

Using (7), (8), and (13), we observe that $\dot{\tilde{\boldsymbol{\theta}}}\to 0$ as $z\to 0$. Since $\ddot{x}_d$ is a bounded function, $\dot{\boldsymbol{\varphi}}\in\mathcal{L}_\infty$. Thus, all the terms in the right-hand side of (20) are bounded, implying that $\ddot{z}\in\mathcal{L}_\infty$. Consequently, $\dot{z}$ is uniformly continuous. Noting that $\int_0^t\dot{z}(v)dv=z(t)-z(0)\to-z(0)$ as $t\to\infty$, and applying Barbalat's lemma, we find that $\dot{z}\to 0$ as $t\to\infty$. Therefore, from (12), $\boldsymbol{\varphi}^T\tilde{\boldsymbol{\theta}}\to 0$ as $t\to\infty$, indicating that the total model compensation error due to parametric uncertainties is asymptotically eliminated by using the parameter adaptation

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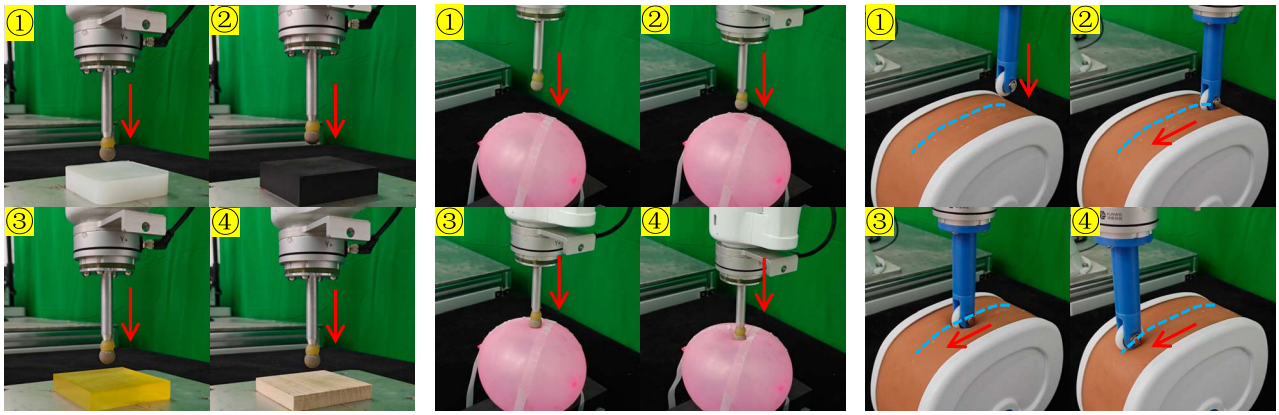


Fig. 2. Experimental setup. (a) Experimental setup of case 1. (b) Experimental setup of case 2 & 3. (c) Experimental setup of case 4.

law in (7), (8), and (13). For any finite T, the fact that $\boldsymbol{\varphi}^T\tilde{\boldsymbol{\theta}} \to 0$ as $t \to \infty$ leads to $\int_t^{t+T} \tilde{\boldsymbol{\theta}}(v)^T \boldsymbol{\varphi}(x(v))\boldsymbol{\varphi}(x(v))^T \tilde{\boldsymbol{\theta}}(v)dv \to 0$ as $t \to \infty$. As $t \to \infty$, since $x(t) \to x_d(t)$ and $\dot{\hat{\boldsymbol{\theta}}} \to 0$ from (7), (8), and (13), the fact that $\int_t^{t+T} \tilde{\boldsymbol{\theta}}(v)^T \boldsymbol{\varphi}(x(v))\boldsymbol{\varphi}(x(v))^T \tilde{\boldsymbol{\theta}}(v)dv \to 0$ is the same as that $\tilde{\boldsymbol{\theta}}(t)^T \int_t^{t+T} \boldsymbol{\varphi}(x_d(v)) \boldsymbol{\varphi}(x_d(v))^T dv \tilde{\boldsymbol{\theta}}(t) \to 0$. Thus, when the PE condition (19) is satisfied, $\tilde{\boldsymbol{\theta}}(t)^T \int_t^{t+T} \boldsymbol{\varphi}(x_d(v))\boldsymbol{\varphi}(x_d(v))^T dv \tilde{\boldsymbol{\theta}}(t) \geq \varepsilon_p ||\tilde{\boldsymbol{\theta}}(t)||^2$, which leads to $\tilde{\boldsymbol{\theta}} \to 0$ as $t \to \infty$. This completes the proof of Theorem 3.

Remark: The above theorems and proofs ensure the stability and convergence of the proposed ARIC. Both the theoretical analysis and practical robotic applications rely on the Assumption [(5) and (6)]. Consequently, in practical implementations, attention should be paid to determining the boundary settings for the structured parameter $\boldsymbol{\theta}$ and the unstructured uncertainty $\tilde{\delta}_1$ based on the specific task scenario.

IV. Experimental Validations

In this section, a seven-DOF robotic manipulator (xMate ER7 pro, ROKAE, China) equipped with a six-axis force/torque (F/T) sensor (KWR75B, Changzhou Kunwei Sensing Technology Co., Ltd., China) mounted on the end-effector is employed to validate the effectiveness of the proposed control scheme. The robotic manipulator operates in the Cartesian position control mode with a control cycle of 1 ms. Furthermore, the nominal resolution of the F/T sensor is 0.06 N and 0.0024 Nm, respectively. The data acquisition frequency of the sensor is 1 KHz. All the algorithms described herein are implemented on a personal computer running the Ubuntu 18.04 operating system, utilizing the RT-Preempt patch to facilitate real-time data exchanges.

As illustrated in Fig. 2, four typical cases are assigned to investigate the performance of different interaction force control approaches. The experiment videos are available at https://youtu.be/Enl0oGE693A.

Case 1: The manipulator contacts surfaces of varying stiffness with a constant desired interaction force.

Case 2: The manipulator contacts a balloon with a time-varying desired interaction force.

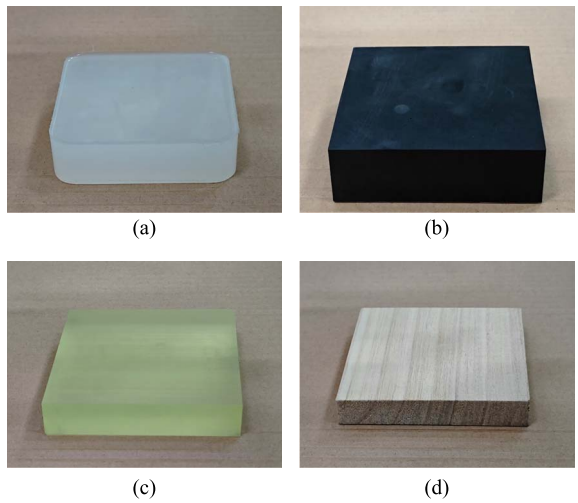


Fig. 3. Contact objects used in the experiments of case 1, with their stiffness increasing from (a) to (d). The Young's moduli of these materials are 0.15426, 4.84294, 8.70295, and 415.25709 MPa, respectively. (a) Soft Silicone. (b) High-density EVA. (c) High-density Polyurethane. (d) Tung Wood.

Case 3: A deviation Δ is deliberately injected into the original command trajectory of the inner-loop motion controller to investigate effects of the inner-loop motion tracking error on the tracking performance of the interaction force. In this way, the actual command trajectory of the inner-loop motion controller becomes $X_m = X_c + \Delta$.

Case 4: The manipulator slides along a given path on a 3-D human abdominal medical phantom while maintaining a constant desired interaction force.

A. Case 1: The Desired Interaction Force Is Constant

In this case, we compare the proposed ARIC with the classical admittance controller (CAC) and the adaptive force-tracking admittance control (AFTAC) proposed by Liu et al. [36]. The AFTAC is based on a disturbance observer to compensate for the geometric and dynamic uncertainties of the environment. To validate the robustness of the ARIC, we utilize various

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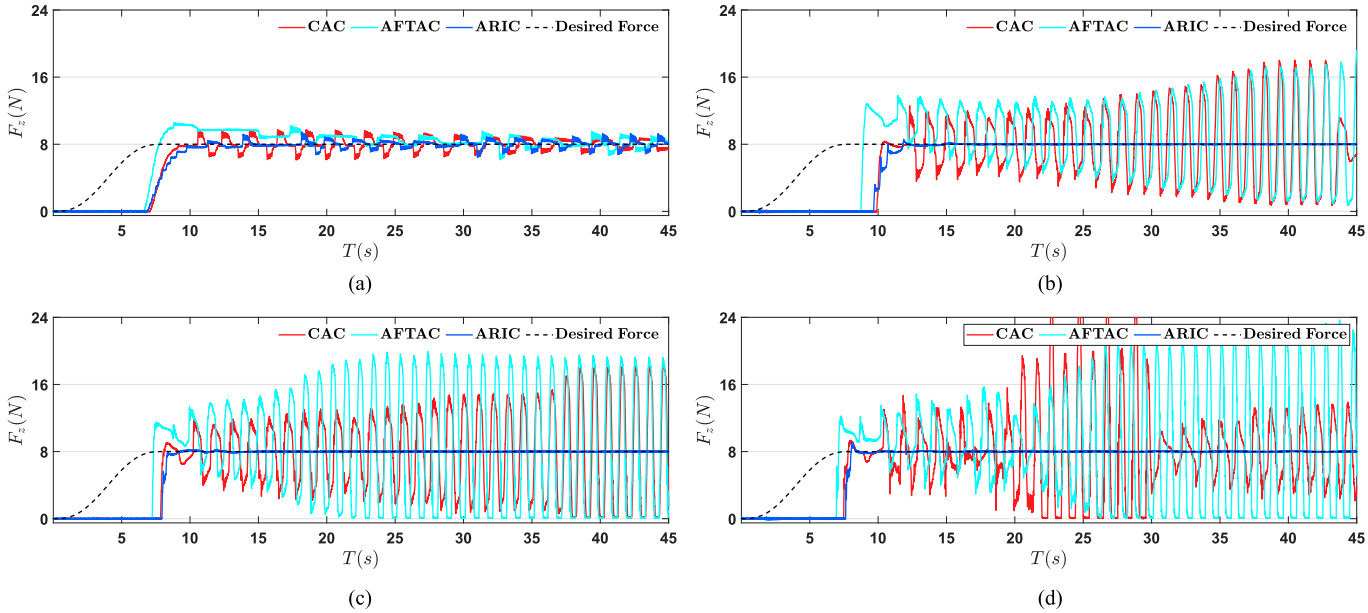


Fig. 4. Interaction force tracking results in case 1. (a) On Soft Silicone. (b) On High-density EVA Material. (c) On High-density Polyurethane. (d) On Tung Wood.

contact materials with different stiffness properties, as depicted in Fig. 3. Specifically, their stiffness increases progressively from Fig. 3(a) to 3(d). As illustrated in Fig. 2, the manipulator is unaware of both the exact environment position and the material stiffness.

Initially, the end-effector of the manipulator is guided to a predetermined point above the object in free space. A desired interaction force of $f_{\text{des}} = 8N$ is set to drive the end-effector downward until a contact is made with the object. A fifth-order polynomial function is employed to ensure a smooth transition of the desired interaction force from zero to the target value over 8 s. Following this, the desired interaction force is expected to be accurately and consistently tracked after a brief transient adjustment. The contact process is segmented into three phases: the approaching stage (free-space), contact transition stage (contact-space) and contact stabilization stage (contact-space). Finally, this process is repeated for all materials with varying stiffness levels.

For equitable comparisons in this case, soft silicone is selected as the reference material. The parameters of ARIC, CAC, and AFTAC are fine-tuned to optimize their respective control performance, resulting in a set of ARIC parameters: $\boldsymbol{\theta}_{\text{min}} = [800,\ -300]^T$, $\boldsymbol{\theta}_{\text{max}} = [1200,\ 300]^T$, $k_1 = 0.15$, $\boldsymbol{\Gamma} = \text{diag}\{50\ 0.5\}$, $\boldsymbol{\theta}_{\text{init}} = [1000,\ 0]^T$, CAC parameters: $M_1 = 1$, $B_1 = 1900$, and AFTAC parameters: $M_2 = 1$, $B_2 = 1900$, $L = 0.05$. These parameters are consistently maintained throughout the experiments in Case 1 for all materials. Additionally, the built-in motion controller of the manipulator is employed as the inner-loop controller.

The results of the comparative experiments conducted on all materials are depicted in Fig. 4. As depicted in Fig. 4(a), it is evident that the steady-state interaction force tracking results of the proposed ARIC, AFTAC, and the CAC fluctuate around the desired value due to the strong elasticity of soft silicone, with the proposed ARIC showing slight superiority over the CAC. AFTAC, however, exhibits a significant overshoot during the transient phase. As commented in [37], admittance-controlled robots may become unstable when in contact with high-stiffness environments, which is confirmed in this experiment as depicted from Fig. 4(b) to 4(d). The red and cyan solid lines in these figures demonstrate that high-amplitude contact oscillations (contact instability) become increasingly prominent as the stiffness of the contact material increases. In fact, due to the static interaction model commonly utilized by admittance-based controllers, environment uncertainties and changes cannot be effectively attenuated. This results in the issue of inevitable force overshooting, consequently limiting their ability to elaborately regulate interaction forces.

In contrast, the blue solid lines in Fig. 4 demonstrate that the proposed ARIC exhibits excellent robustness in maintaining contact stability, even in high-stiffness environments. Taking high-density EVA as an example, the detailed interaction force tracking results and parameter convergence process are depicted in Fig. 5. It clearly demonstrates that from $0 - t_1$, driven by the force tracking error, the end-effector gradually descends in free space and approaches the object to be contacted. The black circles at t_1 signify the initiation of contact. Subsequently, during the $t_1 - t_2$ period, after a brief contact transition adjustment, the actual interaction force smoothly converges to the desired value with a small overshoot. Furthermore, from $t_2 - t_3$, stable contacts are sustained throughout the contact stabilization phase. With the proposed ARIC, the unknown structured parameters $\boldsymbol{\theta} := [\theta_1, \theta_2]^T$ of the developed interaction model are estimated online. These estimated parameters $\hat{\boldsymbol{\theta}} := [\hat{\theta}_1, \hat{\theta}_2]^T$ are then utilized to generate feedforward compensation actions u_a within the ARIC framework, ensuring the exceptional accuracy in the interaction force tracking. Notably, the parameter estimation process consists of several concise equations [specifically (7), (8), and (13)], which simply involve numerical integration to obtain the estimated parameters. This compact formulation

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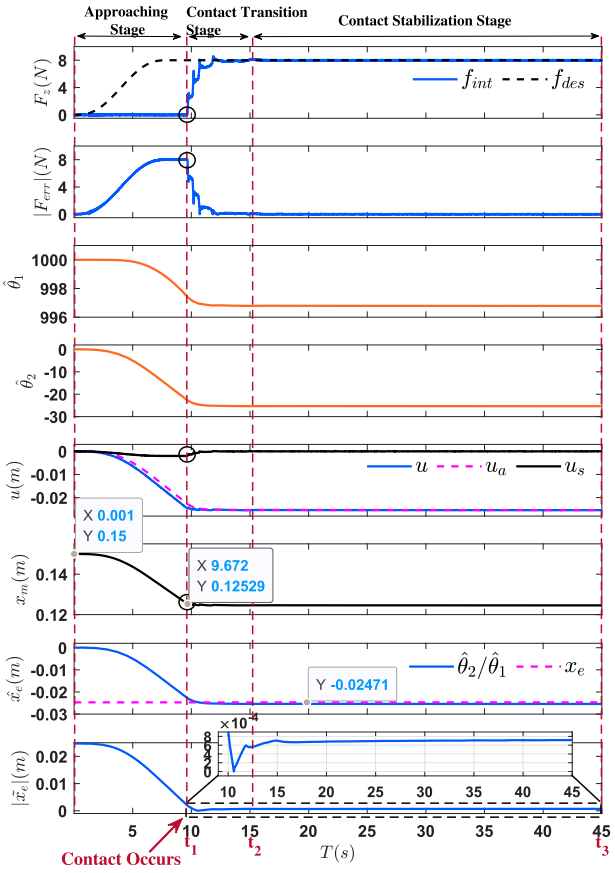


Fig. 5. Parameter convergence process of the proposed ARIC when in contact with high-density EVA.

ensures that the parameter estimation process is computationally efficient and facilitates online adjustment.

B. Case 2: The Desired Interaction Force Is Time-Varying

In this subsection, the desired interaction force is specified as a time-varying trajectory consisting of a quintic polynomial and a sinusoidal function, as formulated as follows:

$$f_{\text{des}} = \begin{cases} \dfrac{20F}{2T^3}t^3 - \dfrac{30F}{2T^4}t^4 + \dfrac{12F}{2T^5}t^5, & \text{if } 0 < t \le T \\ \dfrac{3F}{4} + \dfrac{F}{4} * \sin\left(\dfrac{\pi}{2} + \dfrac{(t-T)}{2}\right), & \text{if } t > T \end{cases} \tag{21}$$

where $F = 8\,\text{N}$, $T = 8\,\text{s}$. Moreover, a fully inflated balloon is selected as the object of contact to evaluate the performance of the proposed ARIC. The specific experimental setup and procedure are depicted in Fig. 2(b). Due to the softness and elasticity of the balloon, its upper surface fluctuates under the press of the end-effector, highlighting the significant position and stiffness uncertainties inherent to the balloon surface. In this case, in addition to the CAC and AFTAC, the adaptive variable impedance controller (AVIC) proposed by Duan et al. [11] is further employed for comparisons. The AVIC adaptively adjusts impedance parameters to mitigate environment position uncertainty.

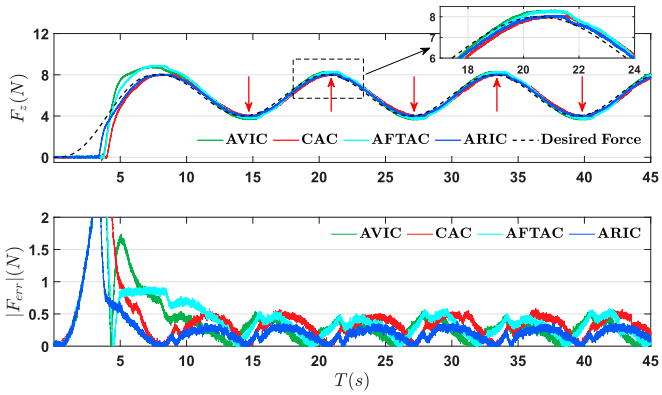


Fig. 6. Interaction force tracking results in case 2.

TABLE I
CONTACT FORCE TRACKING ERRORS OF CASE 2 WITHIN 15 – 45 s

Method	Mean Value [N]	Mean Squared Error [N^2]	Max Value [N]
ARIC (proposed)	**0.187**	**0.043**	**0.383**
AVIC	0.248	0.078	0.547
CAC	0.316	0.120	0.588
AFTAC	0.293	0.107	0.587

Note: The bold entries highlight the performance metrics achieved by the best-performing controller among the four controllers compared.

The experimental results of these controllers are illustrated in Fig. 6. The upper subgraph depicts the actual and desired interaction force curves, while the lower subgraph represents the absolute value of the force tracking errors. From Fig. 6, after contact occurs, the CAC gradually converges to the desired force with a noticeable delay. Once the contact is established, the AVIC and AFTAC quickly tracks the desired interaction force with noticeable overshoot. Furthermore, when the derivative of the desired force changes direction, indicated by the red arrows in Fig. 6, the actual interaction forces of them fail to adjust rapidly enough to accurately track the desired value, resulting in degraded force tracking performance. In contrast, the blue line in Fig. 6 demonstrates that when the proposed ARIC is applied, the actual interaction force rapidly converges to the desired value with only a slight delay after the contact occurs. Furthermore, there is no degradation in tracking accuracy even during significant changes in the curvature of the desired force profile. Additionally, the performance indexes of the interaction force tracking errors are presented in Table I, indicating that the proposed ARIC exhibits minimal dynamic interaction force tracking errors. In terms of MSE, the proposed ARIC controller achieves reductions of 44.9%, 64.2%, and 59.8% compared with AVIC, CAC, and AFTAC, respectively.

C. Case 3: The Inner-Loop Motion Tracking Controller Suffers From Significant Tracking Errors

In this subsection, we investigate the robustness of the interaction force controllers against the inner-loop motion tracking errors. Specifically, following the methodology outlined in the previous subsection, a sinusoidal position deviation signal Δ

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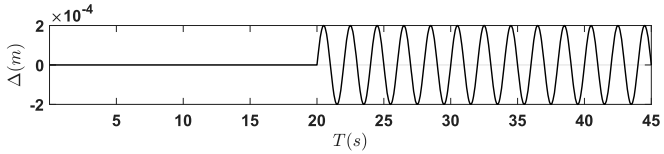


Fig. 7. Deviation signal imposed on the original command trajectory of the inner-loop motion controller.

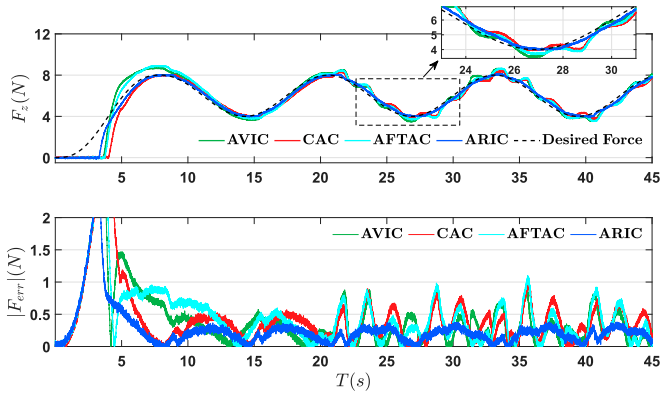


Fig. 8. Interaction force tracking results in case 3.

TABLE II
CONTACT FORCE TRACKING ERRORS OF CASE 3 WITHIN 20−45 s

Method	Mean Value [N]	Mean Squared Error [N^2]	Max Value [N]
ARIC (proposed)	**0.184**	**0.042**	**0.380**
AVIC	0.293	0.135	0.943
CAC	0.338	0.165	0.997
AFTAC	0.322	0.164	1.100

Note: The bold entries highlight the performance metrics achieved by the best-performing controller among the four controllers compared.

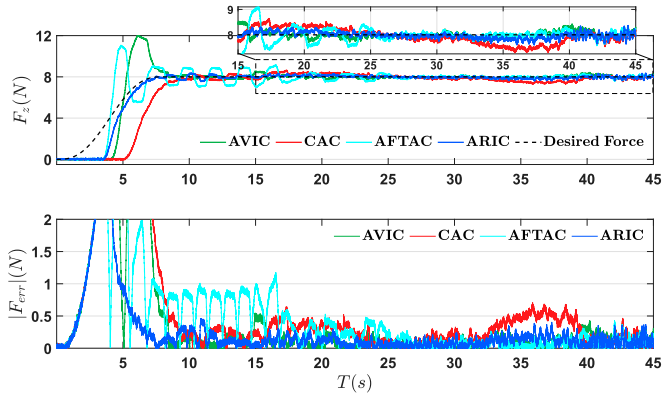


Fig. 9. Interaction force tracking results in case 4.

is added to the outer-loop controller command X_c at 20 s instant, resulting in the alternative command position of the robot being $X_m = X_c + \Delta$. The sinusoidal position deviation signal is specified as $\Delta = 0.0002\sin(2\pi * 0.5t)$, which is depicted in Fig. 7.

The interaction force tracking results of AVIC, CAC, AFTAC, and ARIC are presented in Fig. 8. Before injecting the deviation to the command trajectory of the inner-loop motion controller (prior to 20 s), the four controllers exhibit interaction force tracking performance consistent with Fig. 6. However, once the deviation is introduced (after 20 s), AVIC, CAC, and AFTAC display fluctuations around the desired force, along with a significant increase in force tracking errors. This is due to the introduction of impedance errors whenever the actual position of the robot tip deviates from the commanded value, hindering the accurate realization of the desired impedance dynamics, as discussed in [29], [30]. In contrast, Fig. 8 clearly illustrates that ARIC remains largely unaffected by the command deviation, maintaining a stable and consistent level of interaction force tracking. This robustness of the proposed ARIC is due to our proposed dynamic interaction model (1) capturing unstructured uncertainties such as the inner-loop motion tracking errors intentionally introduced in this experiment. Consequently, these unstructured uncertainties are effectively mitigated by the proposed ARIC. Additionally, Table II presents performance indexes of the interaction force tracking errors with command deviations applied. In terms of MSE, the proposed ARIC achieves reductions of 68.9%, 74.5%, and 74.4% compared with AVIC, CAC, and AFTAC, respectively, which quantitatively confirms the best robust performance of the proposed ARIC.

D. Case 4: Sliding on 3-D Human Abdominal Medical Phantom

In this case, we investigate the controller's performance in regulating interaction force as the robot moves along a predefined path. A 3-D human abdominal medical phantom is specifically selected as the contact object. From 0−15 s, driven by force tracking error, the robot tip moves downward from its predefined position, establishes contact with the abdominal phantom, and gradually reaches the steady state. Between 15−45 s, the robot moves from the starting point to the endpoint along the predefined path while maintaining contact force tracking. This setup effectively simulates basic robotic ultrasound scanning or massage movements.

Notably, the texture of the 3-D abdominal phantom along the path is uneven—some areas feel soft while others feel firm. Additionally, the predefined path on the phantom's surface is not a perfectly horizontal straight line relative to the workbench but rather a curved path with a slight arc. This scenario is particularly valuable for assessing the force controller's ability to adapt to environmental variations and compensate for uncertainties.

As depicted in Fig. 9, the contact force tracking results reveal that during the transient phase of contact, the proposed ARIC quickly and progressively converges to the desired force. The CAC exhibits the slowest convergence speed, while both AVIC and AFTAC show significant overshoots, with AFTAC displaying more pronounced oscillatory behavior. In the steady-state force tracking phase, particularly between 16−24 and 32−40 s, when the abdominal phantom is relatively stiff and soft, respectively, the CAC has difficulty quickly adapting to environmental changes, due to which the contact force is either too large or

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TABLE III
CONTACT FORCE TRACKING ERRORS OF CASE 4 WITHIN 15−45 s

Method	Mean Value [N]	Mean Squared Error [N^2]	Max Value [N]
ARIC (proposed)	**0.102**	**0.016**	**0.403**
AVIC	0.107	0.020	0.538
CAC	0.253	0.091	0.707
AFTAC	0.173	0.070	1.176

Note: The bold entries highlight the performance metrics achieved by the best-performing controller among the four controllers compared.

too small. During the 16−24 s period, AFTAC continues to oscillate around the desired force, with only minimal distinction between ARIC and AVIC. For a clearer comparison, Table III presents the quantitative results of force tracking errors for the compared controllers during the $15-45s$ period. In terms of MSE, the proposed ARIC achieves reductions of 20.0%, 82.4%, and 77.1% compared with AVIC, CAC, and AFTAC, respectively.

V. DISCUSSIONS AND LIMITATIONS

The experiments and results validate the effectiveness of the proposed ARIC method. In uncertain environments, traditional nominal interaction models are unable to adequately capture the interaction dynamics between the robot and the environment. Recognizing these limitations, we propose a novel dynamic interaction model that captures both structured and unstructured uncertainties within the interaction force control framework and synthesize the ARIC to effectively address these uncertainties. By incorporating an adaptive mechanism, unknown structured parameters can be estimated online, leading to an improved feedforward compensation. Additionally, the robust feedback control law in the ARIC can effectively tackle unstructured uncertainties.

The proposed approach can be easily applied in dynamically changing application scenarios, such as robotic ultrasound scanning and physical human–robot interaction (pHRI). For example, when the proposed ARIC is applied in pHRI, the desired interaction force F_d can be specified to zero to minimize the human-robot interaction force. Furthermore, the proposed method operates in the position control mode of the robot with a compact structure and easy deployment. It does not require the availability of an accurate robot dynamic model and the low-level torque control interface, making it applicable to most commercial robotic manipulators.

While our work presents an effective method for controlling robot–environment interaction forces, we recognize certain limitations in its current application. Notably, it relies on an F/T sensor mounted on the end-effector of the robot to measure the interaction force in real time. This dependency increases the implementation costs and reduces the payload capacity of the robotic manipulator. However, these challenges can be overcome by developing sensorless force control approaches. Additionally, synthesizing a high-precision motion controller that can effectively handle both structured and unstructured uncertainties to improve the inner-loop motion tracking accuracy represents a promising direction for future iterations. In our future work, we will further explore the application of the proposed method in real human ultrasound scanning tasks, which involve challenges such as ethical review and human-robot interaction safety.

VI. CONCLUSION

In this article, a novel position-based ARIC scheme is presented along with theoretical stability analysis and experimental validations. In contrast to traditional impedance/admittance-based methods, our approach proposes a novel dynamic interaction model between a robot and the environment that captures both the structured and unstructured uncertainties. The proposed ARIC guarantees precise and robust performance in controlling the interaction forces in uncertain environments. Practical interaction experiments are carried out in different environments. The experimental results confirm that the proposed ARIC has significant advantages over existing typical algorithms in terms of both accuracy and robustness of interaction force control.

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